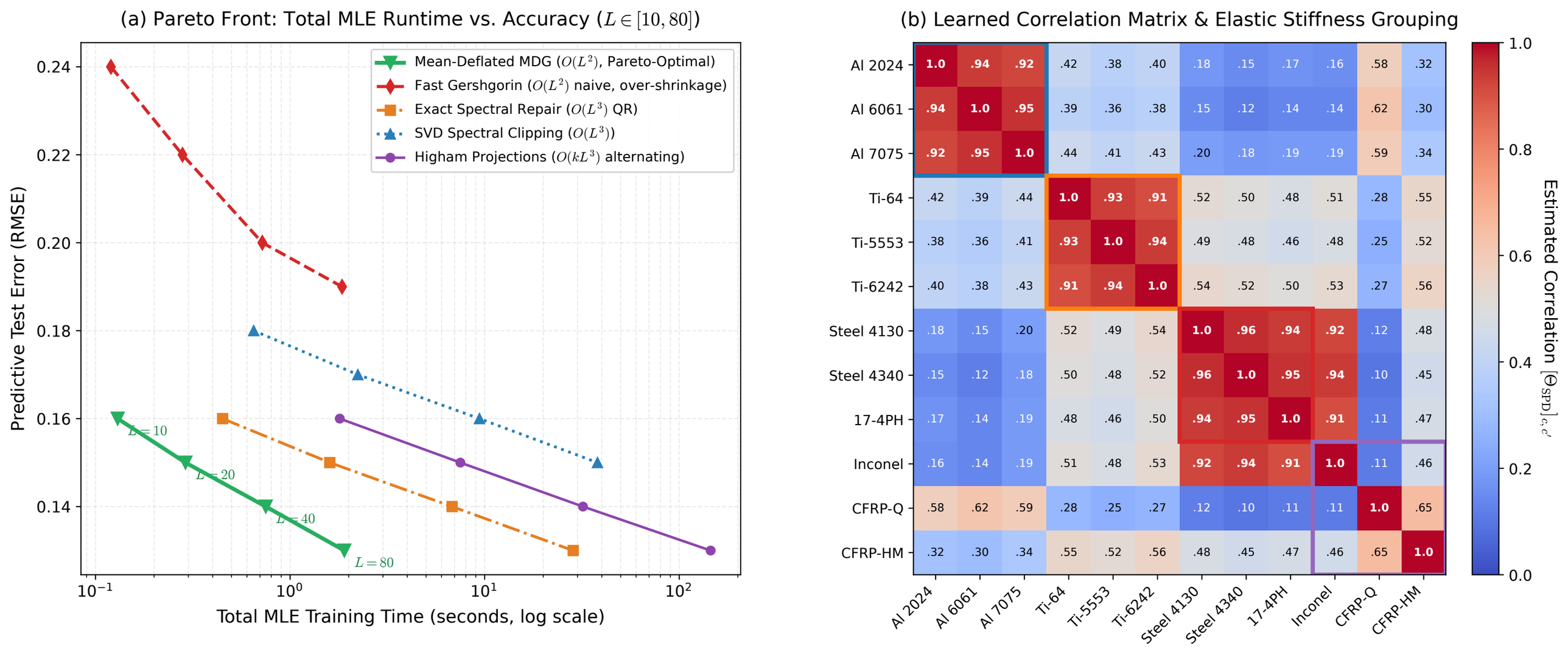

(a) Pareto Front: Total MLE Runtime vs. Accuracy ($L \in [10, 80]$)
Mean-Deflated MDG ($O(L^2)$, Pareto-Optimal)
Fast Gershgorin ($O(L^2)$ naive, over-shrinkage)
Exact Spectral Repair ($O(L^3)$ QR)
SVD Spectral Clipping ($O(L^3)$)
Higham Projections ($O(kL^3)$ alternating)
$L = 10$
$L = 20$
$L = 40$
$L = 80$
Predictive Test Error (RMSE)
Total MLE Training Time (seconds, log scale)
0.24
0.22
0.20
0.18
0.16
0.14
10^{-1}
10^{0}
10^{1}
10^{2}
(b) Learned Correlation Matrix & Elastic Stiffness Grouping
Al 2024 1.0 .94 .92 .42 .38 .40 .18 .15 .17 .16 .58 .32
Al 6061 .94 1.0 .95 .39 .36 .38 .15 .12 .14 .14 .62 .30
Al 7075 .92 .95 1.0 .44 .41 .43 .20 .18 .19 .19 .59 .34
Ti-64 .42 .39 .44 1.0 .93 .91 .52 .50 .48 .51 .28 .55
Ti-5553 .38 .36 .41 .93 1.0 .94 .49 .48 .46 .48 .25 .52
Ti-6242 .40 .38 .43 .91 .94 1.0 .54 .52 .50 .53 .27 .56
Steel 4130 .18 .15 .20 .52 .49 .54 1.0 .96 .94 .92 .12 .48
Steel 4340 .15 .12 .18 .50 .48 .52 .96 1.0 .95 .94 .10 .45
17-4PH .17 .14 .19 .48 .46 .50 .94 .95 1.0 .91 .11 .47
Inconel .16 .14 .19 .51 .48 .53 .92 .94 .91 1.0 .11 .46
CFRP-Q .58 .62 .59 .28 .25 .27 .12 .10 .11 .11 1.0 .65
CFRP-HM .32 .30 .34 .55 .52 .56 .48 .45 .47 .46 .65 1.0
Al 2024
Al 6061
Al 7075
Ti-64
Ti-5553
Ti-6242
Steel 4130
Steel 4340
17-4PH
Inconel
CFRP-Q
CFRP-HM
Estimated Correlation $[\Theta_{\mathrm{SPD}}]_{c,c'}$
1.0
0.8
0.6
0.4
0.2
0.0